

Formalizing my worry

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August 15, 2023

D and U are syntactic variables. $D(n, m)$ represents a number which describes the shape of the m th node in the n th level of the decision tree. $U(n)$ represents the utility our agent gets in round n .¹

Consider the small belief "whenever in $D(n, m)$ Ω asks me for the capital of France and says will give me a dollar if correct, if I answer Paris I receive a dollar". We can formalize it as

$$S \equiv \forall n, m, obs(\phi(D(n, m)) \wedge l(obs) = n \wedge obs_n = D(n, m) \rightarrow (\mathbf{A}(obs) = a \rightarrow U(D(n, m)) = u))$$

Where ϕ checks that the decision node has that syntactic shape, and $l(obs)$ is the length of obs . We could choose more complicated examples, that depend on past or counterfactuals, but let's use this one. We'd like to have something like "if our $\mathbb{Q}$ believes this, then whenever our agent faces such a node, it says Paris".

Say $D(i, j)$ is indeed of that syntactic shape. And i and j are big enough that the sentence explicitly pinning its value, " $D(\underline{i}, \underline{j}) = \underline{D(i, j)}$ ", is large, so that the traders of $\mathbb{Q}$ didn't touch it. Say also obs is the observation sequence up to that point (including $D(i, j)$ as its last observation). Thus their value and relationships to other sentences are set solely through the extrapolation $\mathbb{P}$.

What we want is $\mathbb{P}(U(D(\underline{i}, \underline{j})) = u | \mathbf{A}(\underline{obs}) = a) \approx 1$. More concretely, we want it to be at least as likely as the small belief S . This will be achieved if $\mathbb{P}(U(D(\underline{i}, \underline{j})) = u | S \wedge \mathbf{A}(\underline{obs}) = a) = 1$. This will happen if $\mathbb{P}(\phi(D(\underline{i}, \underline{j})) | S \wedge \mathbf{A}(\underline{obs}) = a) = 1$.²

How do we ensure this? We might be tempted to say "well, probably $\mathbb{Q}$ also knows that all $\underline{i}$ and $\underline{j}$ that look like that satisfy ϕ ". But what do we mean by "look like that"? We mean again something of the form $\phi'(\underline{i}, \underline{j})$. So certainly the extrapolation has to bootstrap us somehow, "noticing that $D(\underline{i}, \underline{j})$ has this property". But it seems like the only way to do it is "computing whether $\phi(D(i, j))$ is true, and including that knowledge in $\mathbb{P}$ (or maybe only inside some of $\mathbb{P}$'s conditionals)".

Now, how do we decide for which $\phi(x)$ to perform this computation (that is, not be updateless)? Of course, just doing all of them would amount to logical updatelessness.

We might say "whenever $\mathbb{Q}$ has a belief of this syntactic form (like S above), then be updateful about ϕ ". This is starting to sound like your Policy selection (but from a fixed-prior, instead of an

¹Not including these changes nothing. This is the same as choosing a concrete syntactic shape for the sentences about the tree and utility that we check.

²Of course we also need $\mathbb{P}(l(\underline{obs}) = n \wedge \underline{obs}_n = D(\underline{i}, \underline{j}) | S \wedge \mathbf{A}(\underline{obs}) = a) = 1$, but let's focus on the above for a moment.

evolving slow LI). And also it's not immediately clear how to make the LI "understand" that this kind of sentences will be used for deciding computation. Our LI is optimizing for epistemic gain, so it will assign probabilities to a sentence of that shape even if it only has a vague guess about its truth. But this will have the "strong" consequence of changing what we are updateful about (and helping set a policy).

One might want to instead "condition in the exact description of $D(i, j)$ when deciding something about an *obs* that includes it". That is, consider

$$\mathbb{P}(U(D(\underline{i}, \underline{j})) = u | S \wedge \mathbf{A}(\underline{obs}) = a \wedge "D(\underline{i}, \underline{j}) = \underline{D(i, j)}")$$

But we face the same problem: $\mathbb{Q}$ didn't have any belief of the form $\phi(\underline{D(i, j)})$, because it's too large, so $\mathbb{P}$ will have to carry out this computation.

Ways forward:

- We do this in an explicit way ("these sentences including this syntactic variable are explicitly to decide what to be updateful about"), and somehow our LI uses them in the right way, "restraining" itself from just getting all the epistemic gains (this is the hard part).
- We stop conditioning on concrete large beliefs, and just condition on properties of different nodes (like $\phi(D(n, m))$). And then, to decide how to act in a concrete node, we somehow average over all the beliefs about properties it satisfies.

Average sounds bad, so instead we maybe should choose between them, but it seems like we don't have a clear criterion to apply. Something like "penalize complexity" seems arbitrary, and something like "keep track of which properties have been useful in the past" is just a restricted kind of Policy Selection.

- Find a syntactical class of ϕ that looks like "the decision-relevant properties"? I don't think this is possible.
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Side-note: Solving this wouldn't nullify my worries about "our frozen prior performing bad in the limit". Even if there is "a finite set of properties" that $\mathbb{Q}$ had opinions on and thus we always check to make decisions, there will be infinitely many we're missing, and they could be arbitrarily important in the future.

Another angle: If we were just continuously running an LI, we'd eventually have good guesses about which concrete numerals satisfy which properties. But given a prior into which only finitely much computation has gone, there will of course be infinitely many numerals about which we have no idea about their properties. So this feels like pointing towards "you always need some kind of computation, and have to (dynamically?) decide which things to compute".

Actually, the situation is subtler: it's not true that only finitely much computation has gone into $\mathbb{P}$. Sure, $\mathbb{Q}$ is finite, but computing $\mathbb{P}(\phi_n)$ requires arbitrary computation for arbitrary n . The thing is we are not dynamically deciding which things to compute: we fixed at the start how to compute all $\mathbb{P}$. The vibe I've been getting is the following:

- If you fix these computations to be narrow (and only, for example, replace some things by others that $\mathbb{Q}$ already knows are equivalent), then you still get the problem explained above.
- If you fix these computations to be general enough to propagate beliefs as we need above, then you are dealing with general enough computation (computing whether $\phi(D(i,j))$) that you need to "manually narrowly restrict it" (by saying something like "only do this for the ϕ in this class"). And a priori it seems like this will just be fixed-prior Policy Selection, but I need to explore it more.